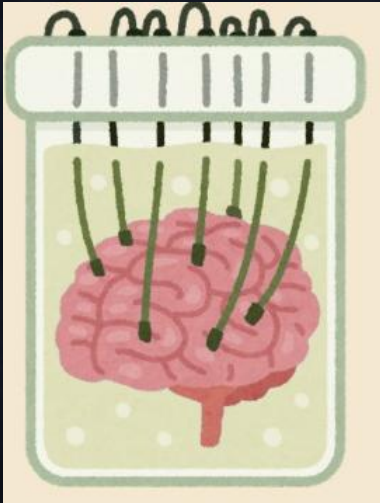


A New Age of Consciousness Research:
Novel Techniques for a Millennia-Old Question

Hunor Bartalis-Szélyes
June, 2025

The beginnings



"Brain in a vat" by Neurarcane ^[1]

Who are we, really?
Where does our experience of the
world reside?
And how could we ever hope to study
it scientifically?

"William & Mary" (1954)



Summary

1. What is consciousness?
2. Brief (and incomplete) history of consciousness research
3. The “hard problem of consciousness”
4. Learning from history
5. Neural correlates of consciousness (NCCs)
6. Emerging techniques to study NCCs
7. Ending

1. So, what is consciousness?

Join at menti.com | use code 4805 7033

Mentimeter

So, what is consciousness?



👍 👤

2. Brief (and incomplete) history of studying consciousness

~13000–3000 BCE

Prehistoric roots

Earliest attempts to manipulate or alter our subjective experience
by interfering with brain functioning

Beer (~13.000 BCE in Natufian culture, Israel)

San Pedro (~8600 BCE in Peru)

Hemp (~8200 BCE in Jomon culture, Japan)

Psilocibin (~6000 BCE in Sahara Desert & ~ 4000 BCE, Spain)
and more...^[a]

Trephination



Image by Dani Nadel, [CC BY 3.0](#), via Wikimedia Commons ^[2]



Image by Thermochopps, [CC BY-SA 4.0](#), via Wikimedia Commons ^[3]



Image by United States Fish and Wildlife Service, Public domain, via Wikimedia Commons ^[4]



Image by Vincent Verroust, [CC BY-SA 4.0](#), via Wikimedia Commons ^[5]



Image by Silar, [CC BY-SA 4.0](#), via Wikimedia Commons ^[6]

2. Brief (and incomplete) history of studying consciousness

~13.000–3000 BCE



Image by Jeff Dahl, Public domain, via
Wikimedia Commons ^[7]

~ 3000-1600 BCE

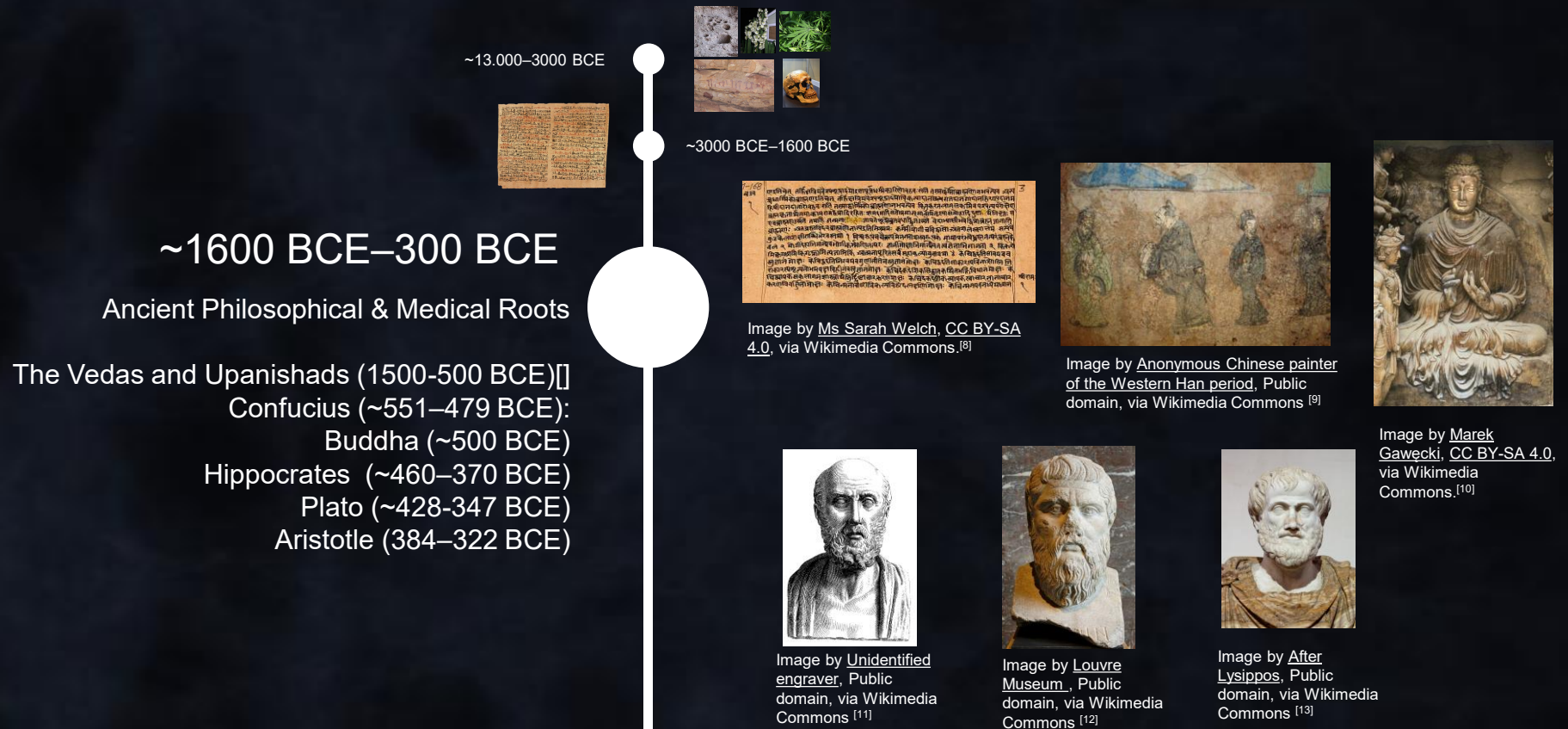
Bronze Age proto-neurology

Edwin Smyth Papyrus

Unknown Egyptian physicians

Earliest known empirical/medical case
descriptions linking brain injuries to impairments
in the experience of humans

2. Brief (and incomplete) history of studying consciousness

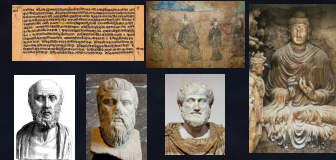


2. Brief (and incomplete) history of studying consciousness

~13.000–3000 BCE



~3000 BCE–1600 BCE



~1600 BCE–300 BCE



Image by [Philippe de Champaigne](#), Public domain, via Wikimedia Commons ^[14]



Image by [Shouyi 首一](#), Public domain, via Wikimedia Commons ^[15]



Image by [Beckwith](#), Public domain, via Wikimedia Commons ^[16]

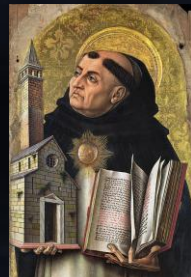


Image by [Carlo Crivelli](#), Public domain, via Wikimedia Commons ^[17]



Image by [Vicente Salvador Gómez](#), Public domain, via Wikimedia Commons ^[18]

~300 BCE–1500 CE

Preservation and Theological Integration

Augustine of Hippo (354 CE - 430 CE)

Vasubandhu (4th century CE – 5th century CE)

Ibn Sina (980 CE – 1037 CE)

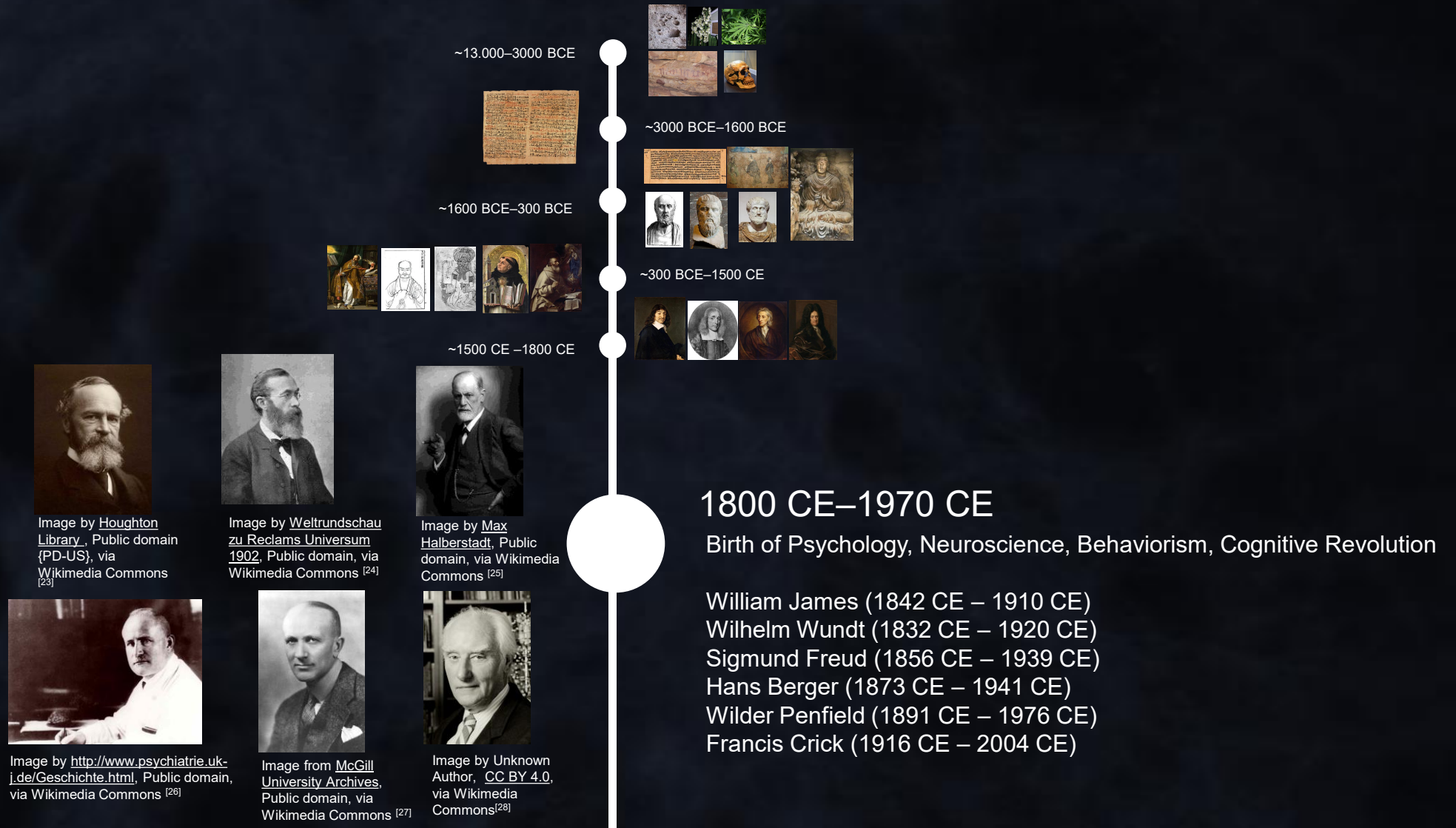
Thomas Aquinas (1225 CE – 1274 CE)

Albertus Magnus (1200 CE – 1280 CE)

2. Brief (and incomplete) history of studying consciousness



2. Brief (and incomplete) history of studying consciousness



2. Brief (and incomplete) history of studying consciousness



1970 CE–Present

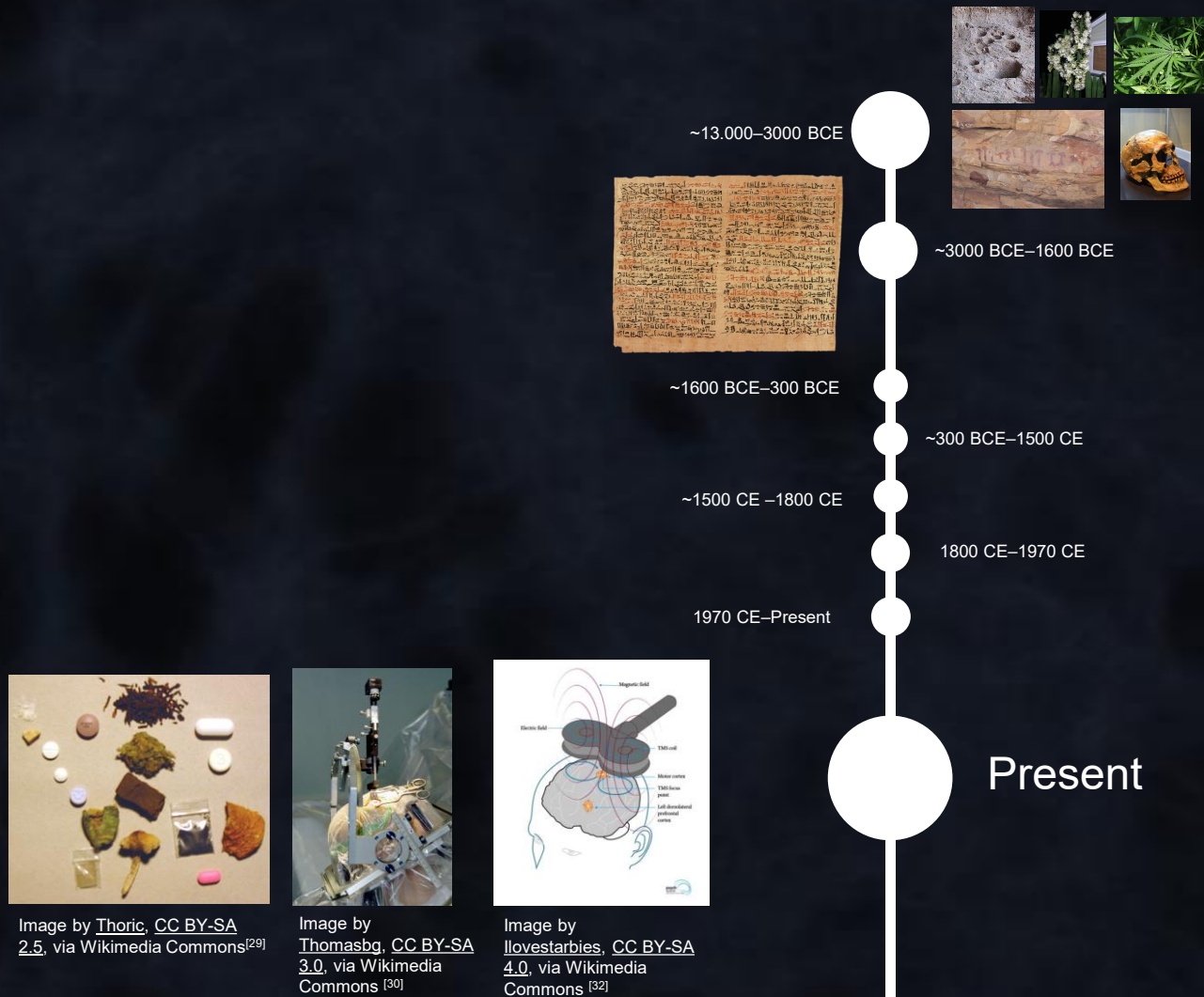
Neuroscientific and Computational Consciousness Research

Daniel Dennett (1942 CE - 2024 CE)
Christof Koch (1956 CE - Present)
Thomas Metzinger (1958 CE - Present)
David Chalmers (1966 CE – Present)
Giulio Tononi (1960 CE - Present)
Stanislas Dehaene (1965 CE - Present)
Anil Seth (1972 CE - Present)
and perhaps you will be the next on this list



Image by [Vector Portal](#), CC BY 4.0,
via [Wikimedia Commons](#)¹

2. Brief (and incomplete) history of studying consciousness



2. Brief (and incomplete) history of studying consciousness

“But fundamentally an organism has conscious mental states if and only if there is something that it is like to be that organism – something it is like for the organism. We may call this the subjective character of experience. (Nagel, 1974, p.436) ^[b]

“If physicalism is to be defended, the phenomenological features must themselves be given a physical account. But when we examine their subjective character it: seems that such a result is impossible. The reason is that every subjective phenomenon is essentially connected with a single point of view, and it seems inevitable that an objective, physical theory will abandon that point of view.” (Nagel, 1974, p.437) ^[b]

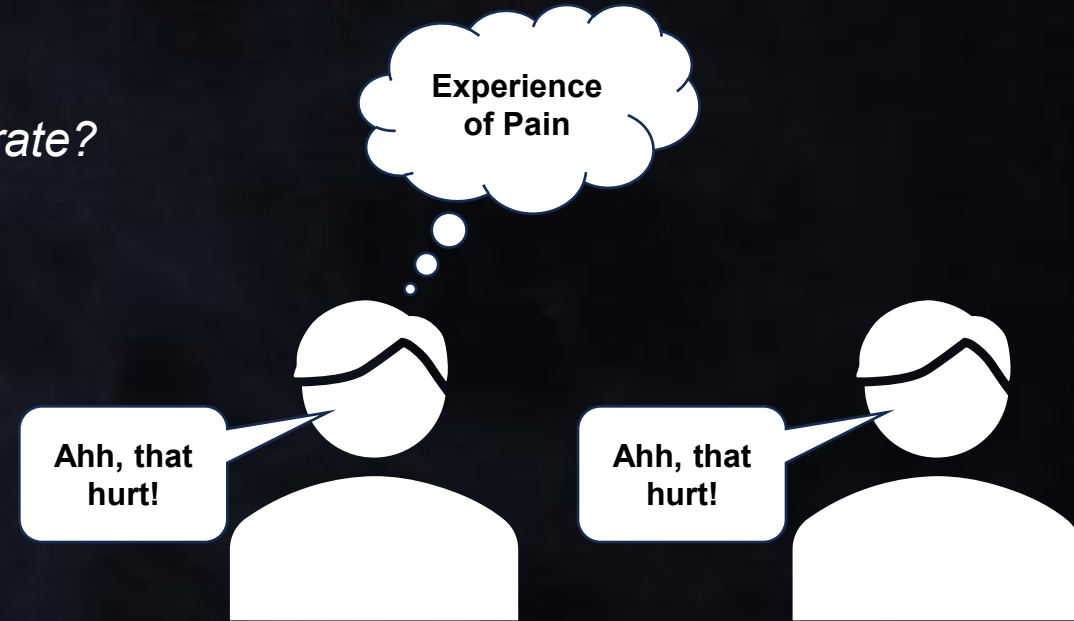
3. The “hard problem of consciousness”

How can subjective experience arise from a physical substrate?

“Mill Argument” by Leibniz (1714) [c]

“Imitation man” argument by Keith Campbell (1970) [d]

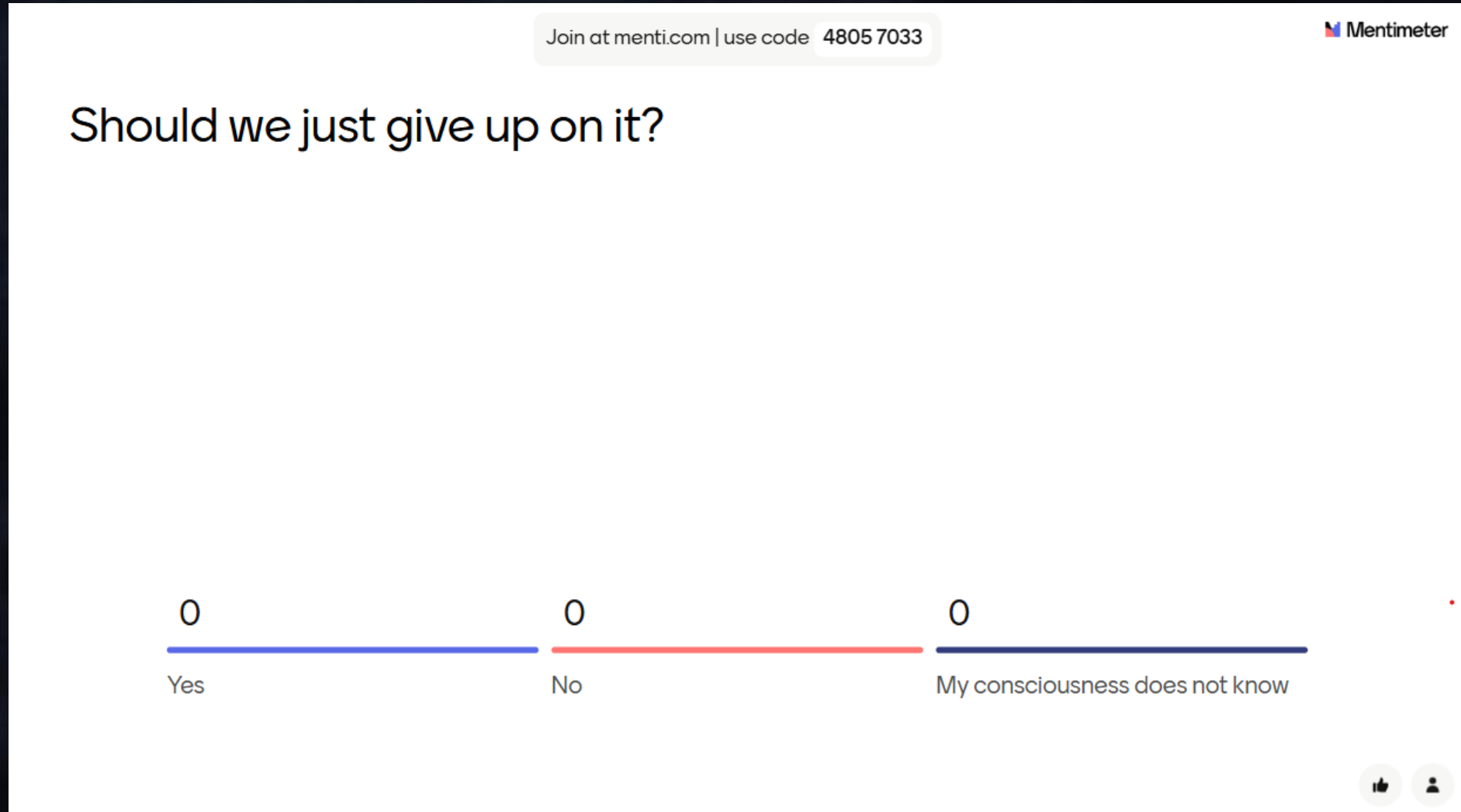
“Philosophical zombie” by David Chalmers (1996) [e]



3. Are we just philosophical zombies or there is more to consciousness?



3. Should we just give up on it?



3. Should we just give up on it?

Theoretical

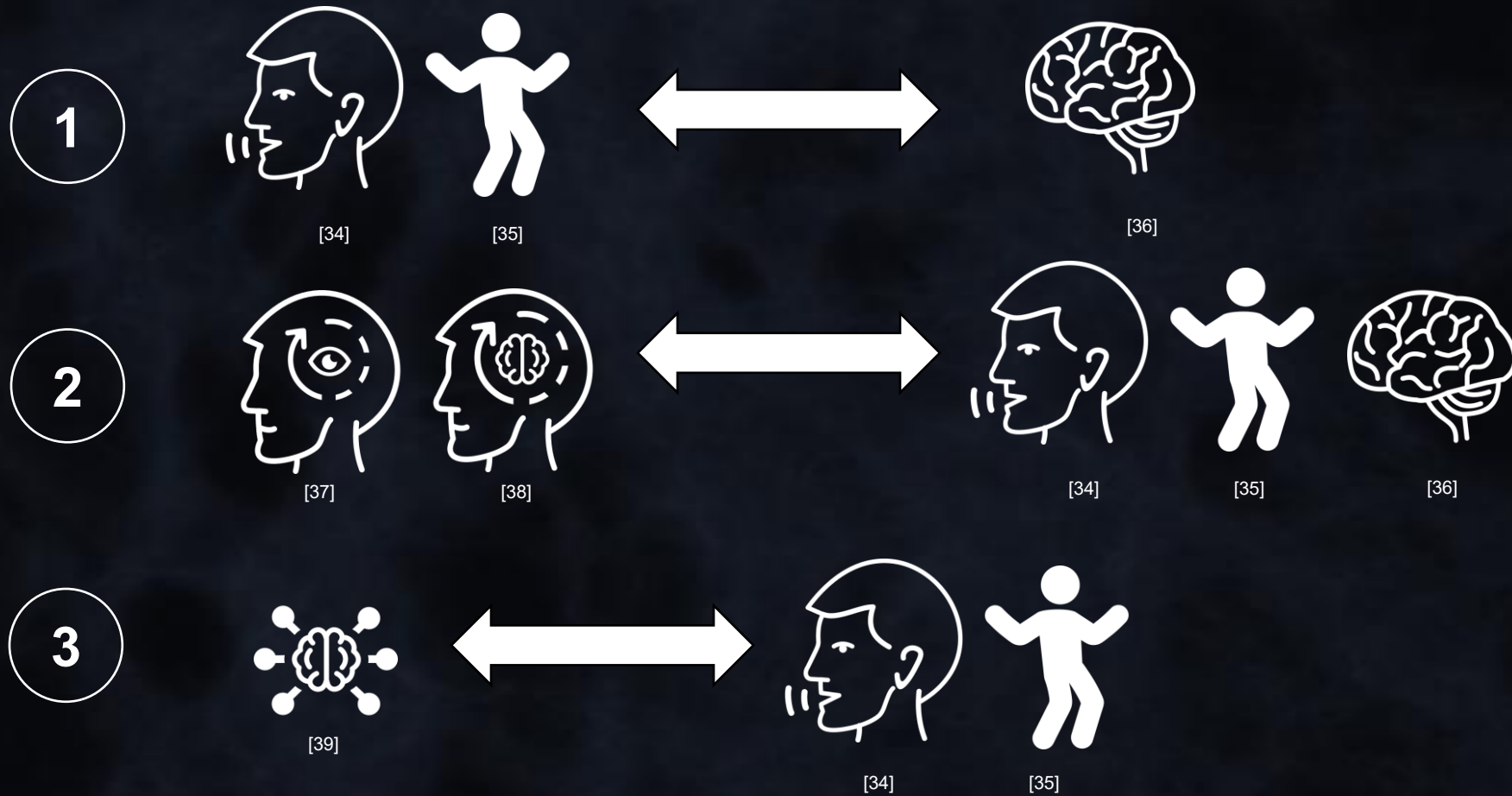


Image by [RANGGI MANGGALA ranggimr](#), [CC0 1.0](#), via Wikimedia Commons ^[33]

Clinical

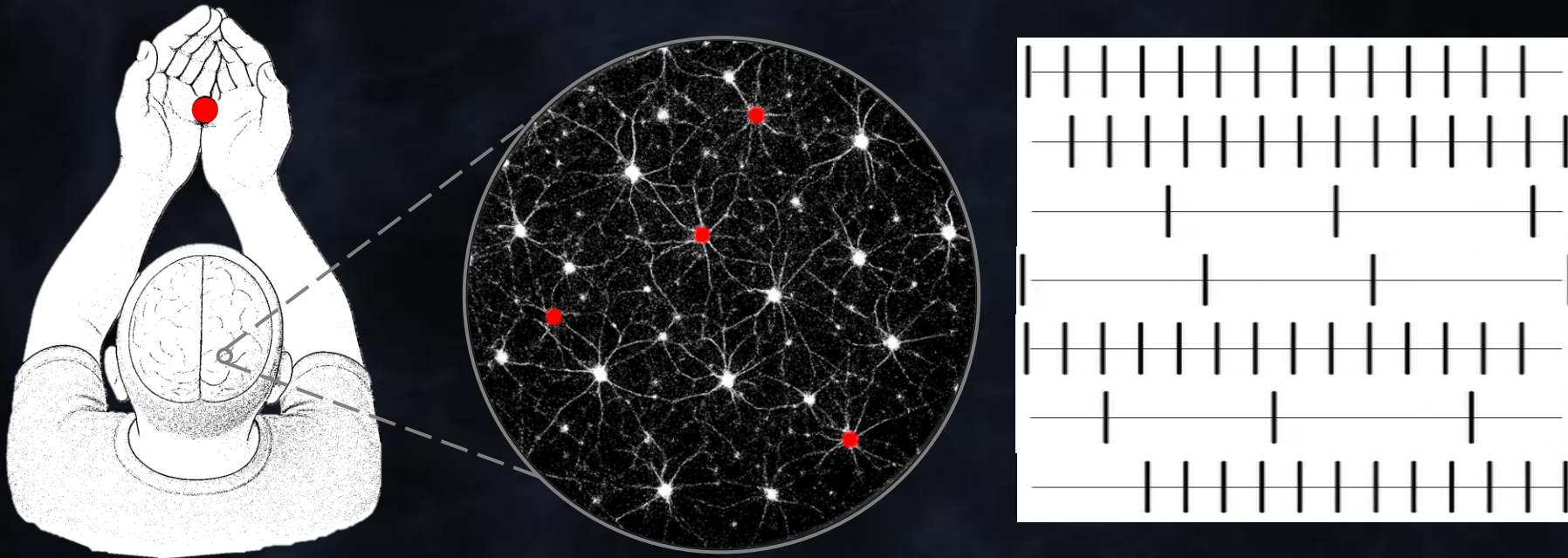
DEREALIZATION
VEGETATIVE STATE
PSYCHOSIS
LOCKED-IN SYNDROME
GENERAL ANAESTHESIA
COMA
LUCID DREAMING
MINIMALLY CONSCIOUS STATE
HALLUCINATIONS
DEPERSONALIZATION

4. Learning from history



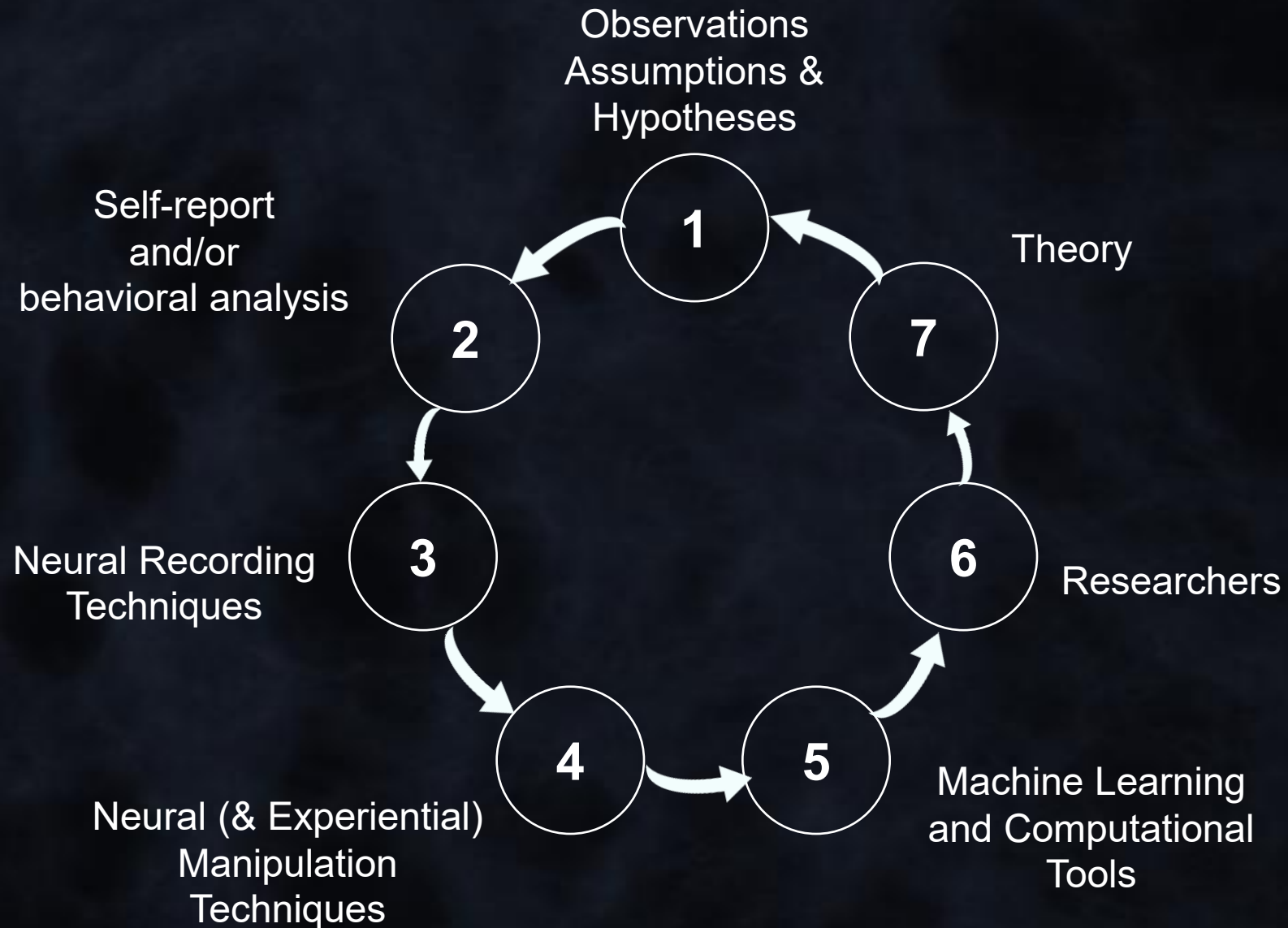
5. Neural correlates of consciousness (NCC) and beyond

NCCs are “the minimum neuronal mechanisms jointly sufficient for any one specific conscious experience” [f]



These images were created with the assistance of DALL-E 3

What would we need?



1. Observations, assumptions, & hypotheses

1. Consciousness exists (whatever it may be)
2. Humans (and arguably many animals) have a subjective experience
3. The self-reports of others relates to conscious experience (even if indirectly)
4. Some behaviors of others relates to conscious experience (even if indirectly)
5. The brain's activity relates to conscious experience (even if indirectly)
6. We can operationalize and measure the self-reports, behaviors, and brain activity

and the list of assumptions could go on...

2. Self-report and/or behavioral analysis (especially in other animals)

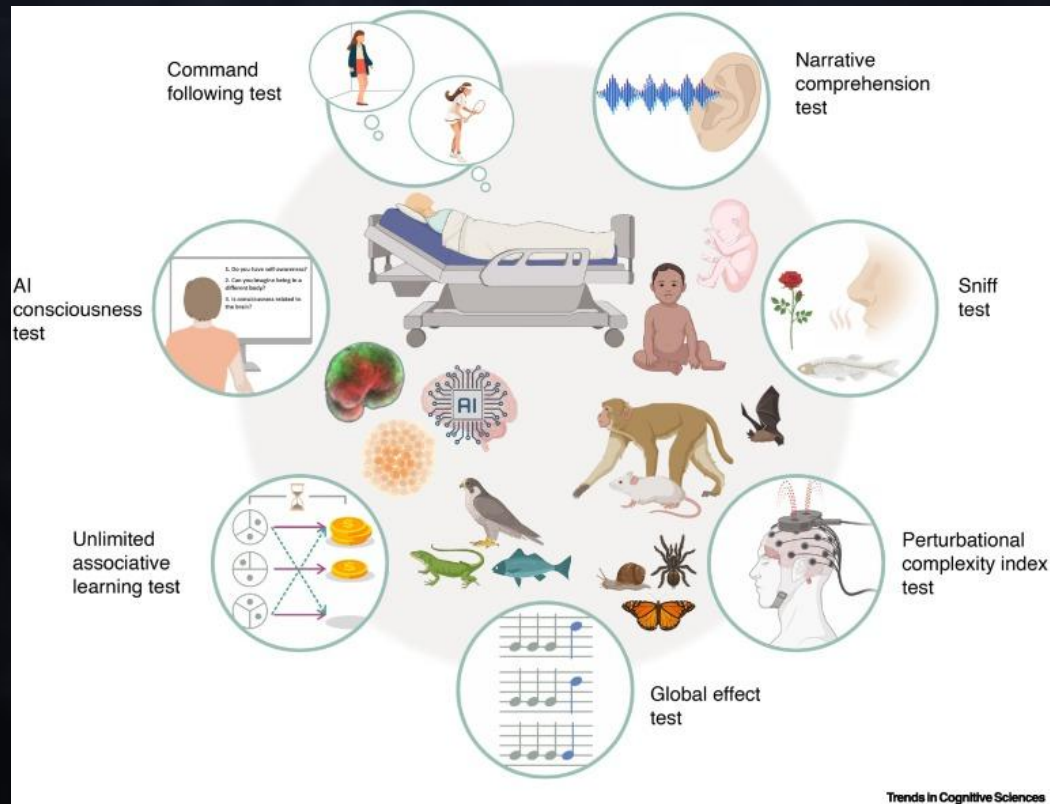


Figure 1 Examples of proposed tests for consciousness (C-tests) from:

Bayne, T., Seth, A. K., Massimini, M., Shepherd, J., Cleeremans, A., Fleming, S. M., Malach, R., Mattingley, J. B., Menon, D. K., Owen, A. M., Peters, M. A., Razi, A., & Mudrik, L. (2024). Tests for consciousness in humans and beyond. *Trends in Cognitive Sciences*, 28(5), 454–466. <https://doi.org/10.1016/j.tics.2024.01.010>

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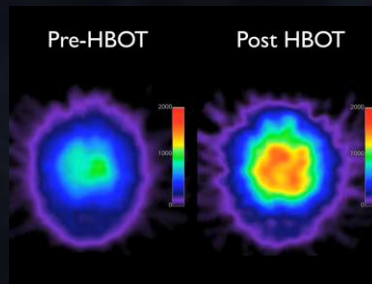
What do you think could be a good way to assess self-reports of subjective experience?

Which behaviors do you think are indicative of conscious experience in humans? How about animals?

3. Neural Recording Techniques Before

Indirect

SPECT



Video by [Harch P, Fogarty E, Staab P, Van Meter K](#), CC BY 3.0, via Wikimedia Commons [40]

PET

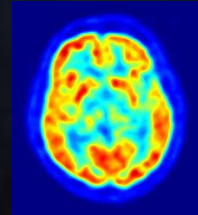
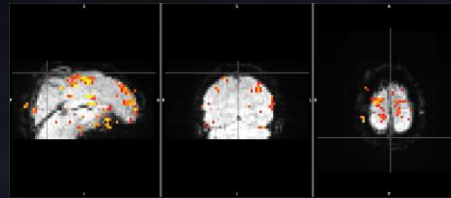


Image by [Juan galan](#), CC BY-SA 3.0, via Wikimedia Commons[41]

(f)MRI



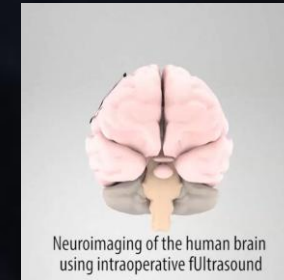
Video by [Tong Y, Hocke L, Fan X, Janes A, Frederick B](#), CC BY 4.0, via Wikimedia Commons [42]

fNIRS



Image by [Mentorklaspsy](#), CC BY-SA 4.0, via Wikimedia Commons [43]

fUS



Video S1 from:
Imbault, M., Chauvet, D., Gennisson, JL. *et al.* Intraoperative Functional Ultrasound Imaging of Human Brain Activity. *Sci Rep* 7, 7304 (2017).
<https://doi.org/10.1038/s41598-017-06474-8>
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3. Neural Recording Techniques Before

Single-unit microelectrode recordings

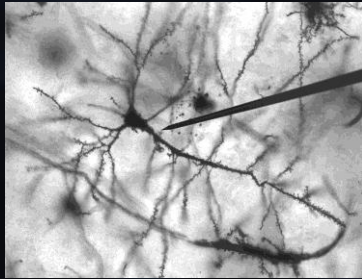
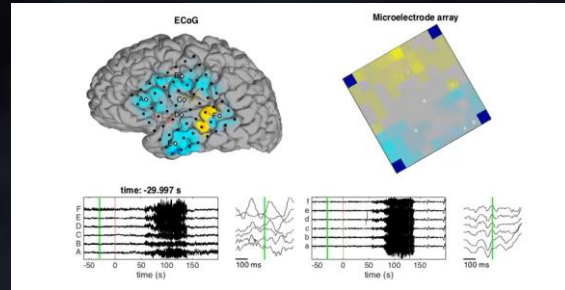


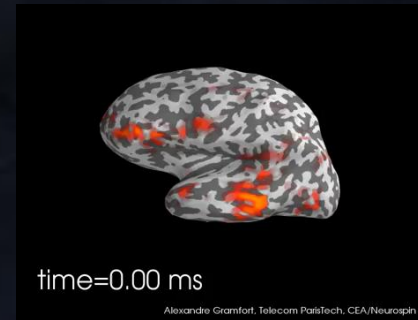
Image by [Laurens R. Krol](#), [CC BY-SA 4.0](#), via Wikimedia Commons [44]

Intracranial EEG/ECoG Multi-unit recordings



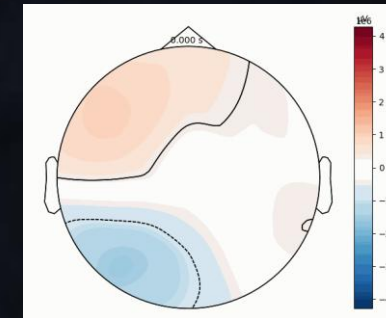
Video by [Martinet L.](#), [Fiddymont G.](#), [Madsen J.](#), [Eskandar E.](#), [Truccolo W.](#), [Eden U.](#), [Cash S.](#), [Kramer M.](#), [CC BY 4.0](#), via Wikimedia Commons [45]

MEG



Video by [Alexandre Gramfort](#), [CC BY 3.0](#), via YouTube [46]

Surface EEG



Direct
(electro-
magnetic)

3. Neural Recording Techniques Today – Neuropixels

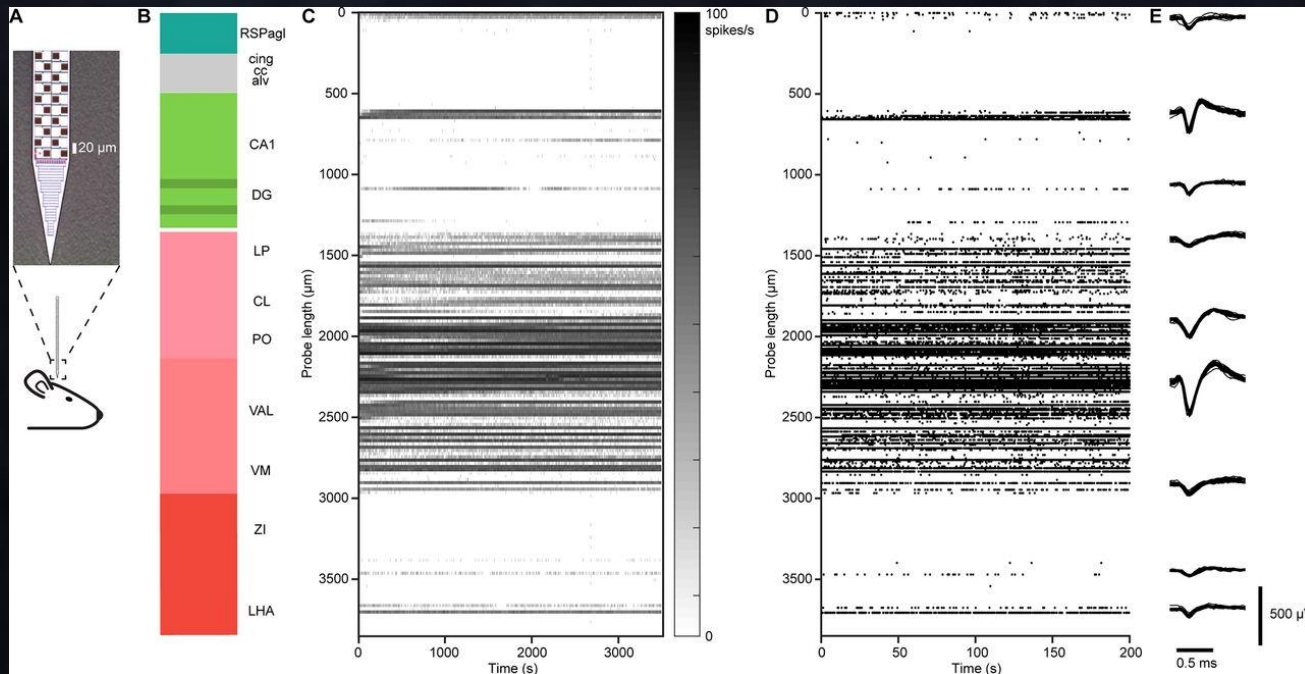
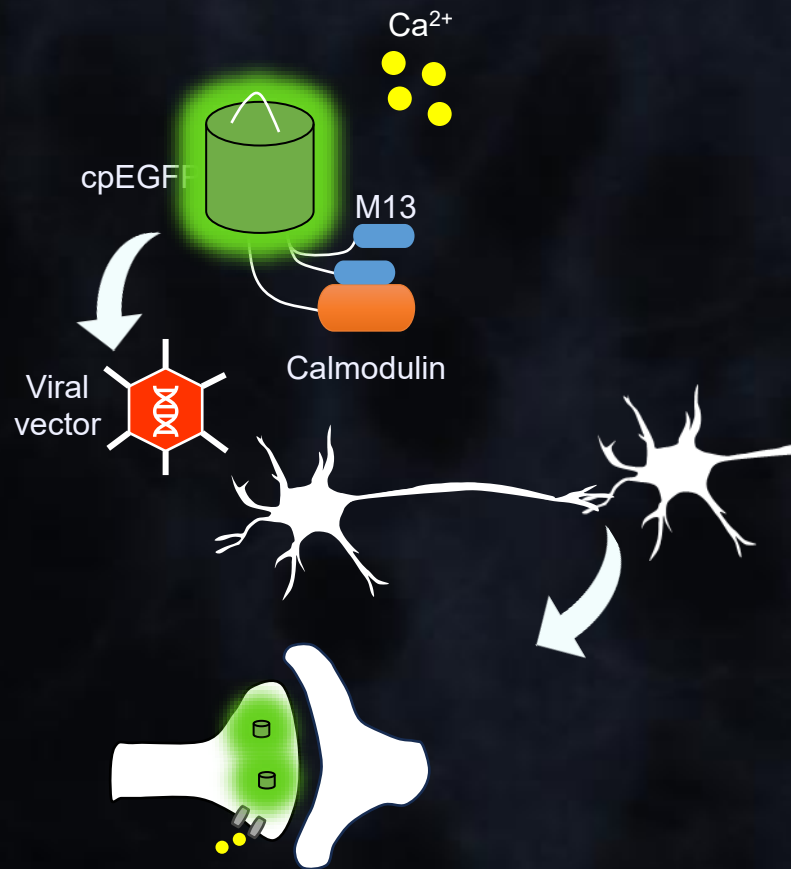


Figure 2. Neuropixels probe recording from:

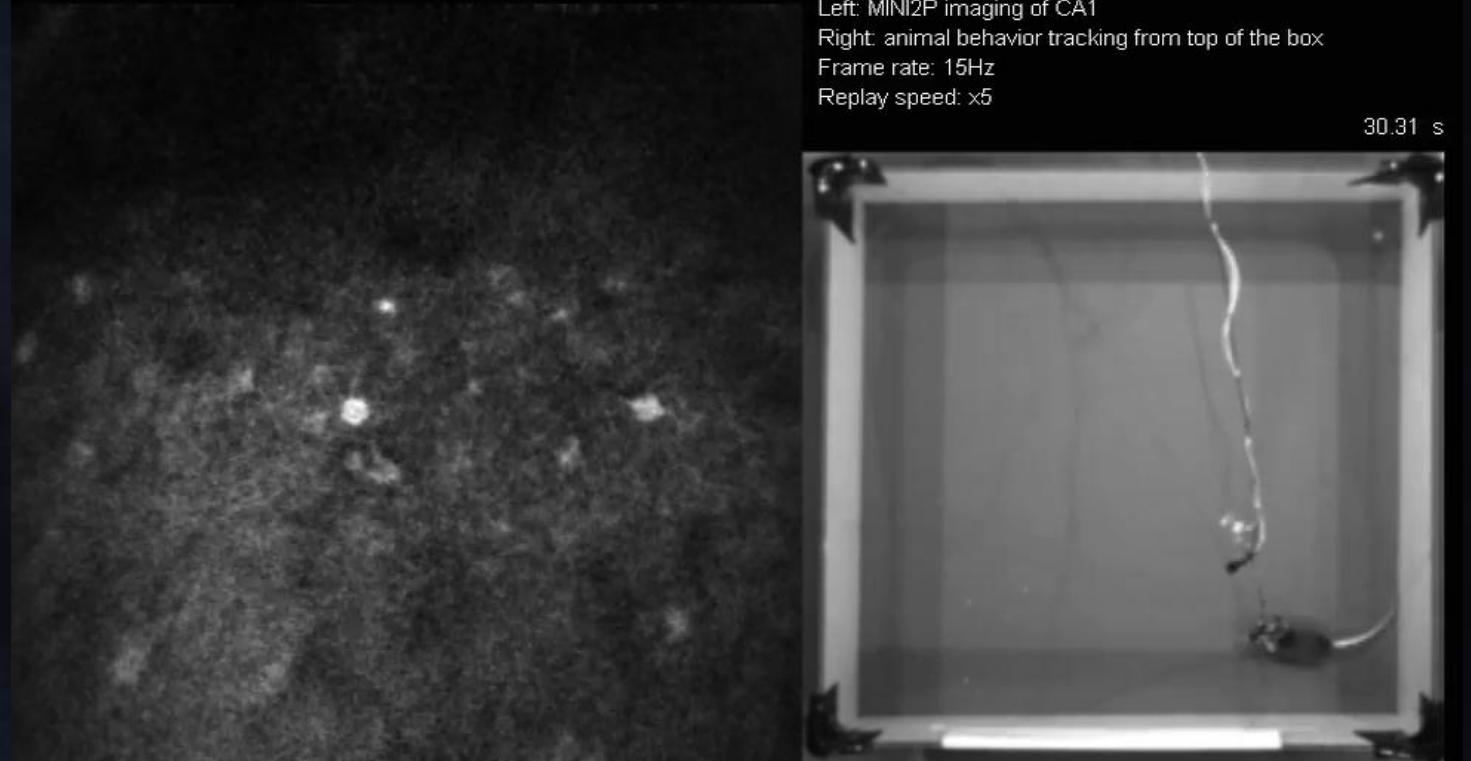
Liu, L. D., Chen, S., Hou, H., West, S. J., Faulkner, M., Economo, M. N., Li, N., & Svoboda, K. (2021). Accurate Localization of Linear Probe Electrode Arrays across Multiple Brains. *eNeuro*, 8(6), ENEURO.0241-21.2021. <https://doi.org/10.1523/eneuro.0241-21.2021>
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In humans:
Paulk et al. (2022)[g]
Coughlin et al. (2023)[h]

3. Neural Recording Techniques Today – Calcium Imaging



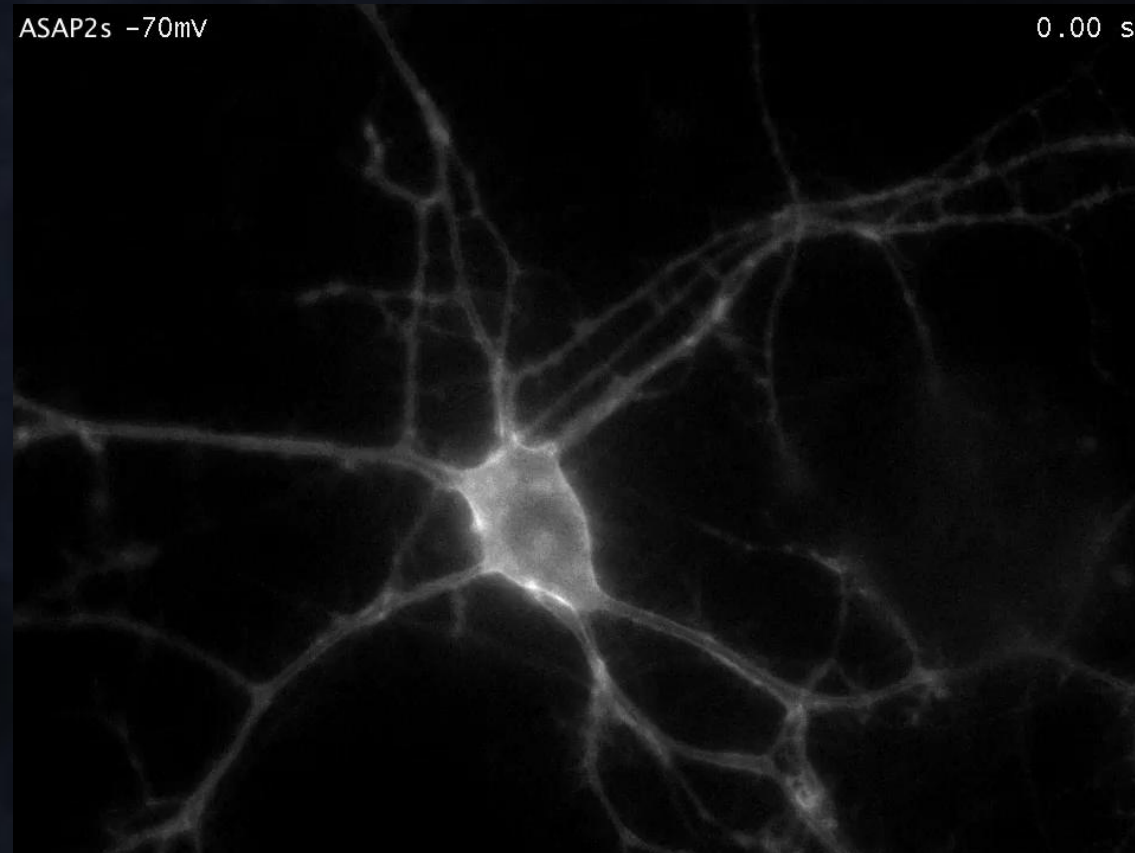
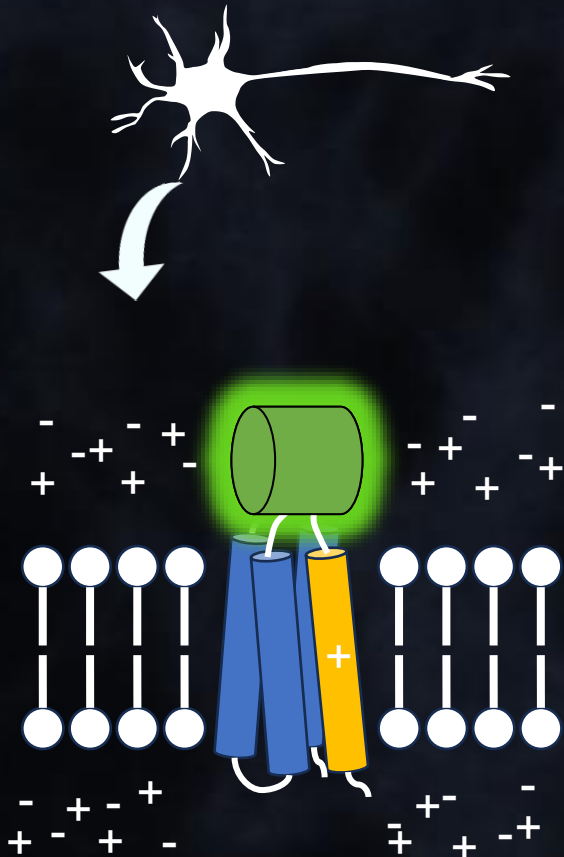
Part III: example imaging of calcium activity in hippocampal area CA1 during unrestrained movement



Left: MINI2P imaging of CA1
Right: animal behavior tracking from top of the box
Frame rate: 15Hz
Replay speed: x5

Video S1 from:
Zong, W., Obenaus, H. A., Skytøen, E. R., Eneqvist, H., De Jong, N. L., Vale, R., Jorge, M. R., Moser, M.-B., & Moser, E. I. (2022). *Large-scale two-photon calcium imaging in freely moving mice*. *Cell*, 185(7), 1240–1256.e30. <https://doi.org/10.1016/j.cell.2022.02.017>
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3. Neural Recording Techniques Today – Voltage Imaging



Video 2 from:

Chamberland, S., Yang, H. H., Pan, M. M., Evans, S. W., Guan, S., Chavarha, M., Yang, Y., Salesse, C., Wu, H., Wu, J. C., Clandinin, T. R., Toth, K., Lin, M. Z., & St-Pierre, F. (2017). Fast two-photon imaging of subcellular voltage dynamics in neuronal tissue with genetically encoded indicators. *eLife*, 6. <https://doi.org/10.7554/elife.25690>

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4. Neural (and Experiential) Manipulation Techniques Before

Microelectrodes
Deep Brain Stimulation
Transcranial magnetic stimulation
Transcranial direct-current stimulation
Transcranial focused ultrasound stimulation



Image by [Jmarchn](#), [CC BY-SA 3.0](#), via Wikimedia Commons [47]

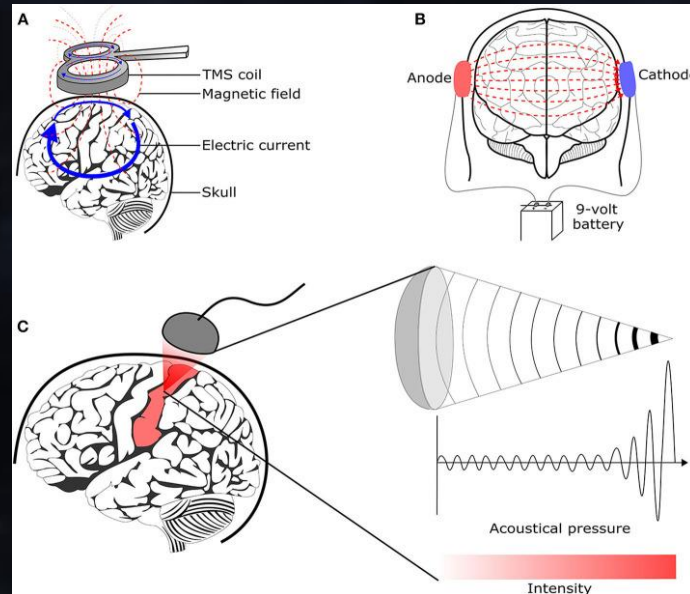


Image by [Taeho Kim et al.](#), [CC BY 4.0](#), via Wikimedia Commons [48]
(Kim et al., 2020)

Sensory stimulation



Image by [Allan-Hermann Pool](#), [CC BY-SA 4.0](#), via Wikimedia Commons [49]

4. Neural (and Experiential) Manipulation Techniques Today – Psychoactive Substances



Image by [Jynto](#) (more from this user),
CC0, via Wikimedia Commons [50]

MDMA
(3,4-
Methylenedioxymetha
mphetamine)

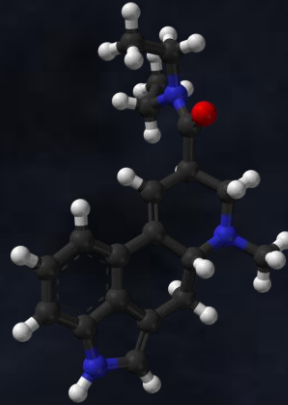


Image by [Benjah-bmm27](#), Public
domain, via Wikimedia Commons [51]

LSD
(Lysergic
acid
diethylamide)

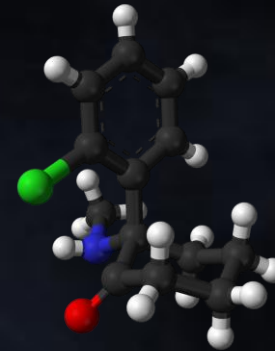
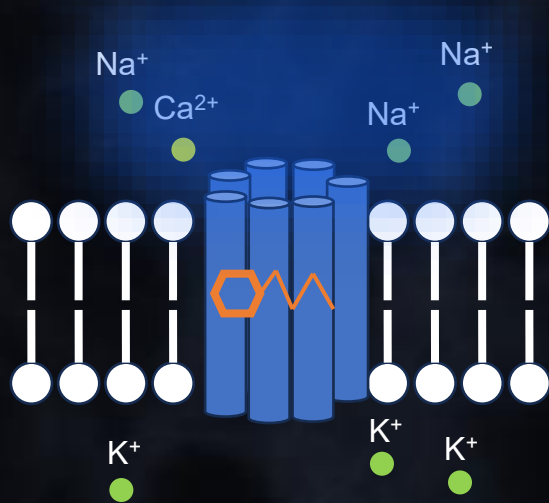


Image by [Benjah-bmm27](#), Public
domain, via Wikimedia Commons [52]

Ketamine

4. Neural (and Experiential) Manipulation Techniques Today – Optogenetics



In humans:
Sahel et al. (2021)

Video by [Wei P, Liu N, Zhang Z, Liu X, Tang Y, He X, Wu B, Zhou Z, Liu Y, Li J, Zhang Y, Zhou X, Xu L, Chen L, Bi G, Hu X, Xu F, Wang L](#), CC BY 4.0, via Wikimedia Commons []

Wei, P., Liu, N., Zhang, Z., Liu, X., Tang, Y., He, X., Wu, B., Zhou, Z., Liu, Y., Li, J., Zhang, Y., Zhou, X., Xu, L., Chen, L., Bi, G., Hu, X., Xu, F., & Wang, L. (2015). Processing of visually evoked innate fear by a non-canonical thalamic pathway. *Nature Communications*, 6(1). <https://doi.org/10.1038/ncomms7756>

5. Machine Learning & Computational Models

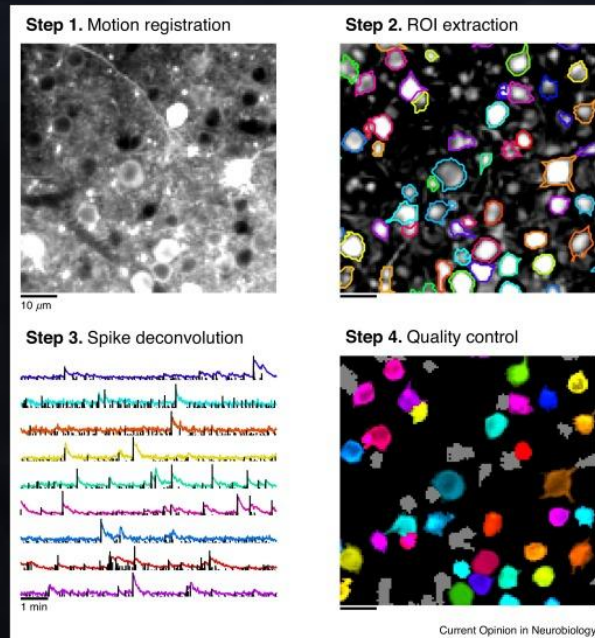


Figure 1. Pipeline for calcium imaging from: Stringer, C., & Pachitariu, M. (2018). Computational processing of neural recordings from calcium imaging data. *Current Opinion in Neurobiology*, 55, 22–31. <https://doi.org/10.1016/j.conb.2018.11.005>

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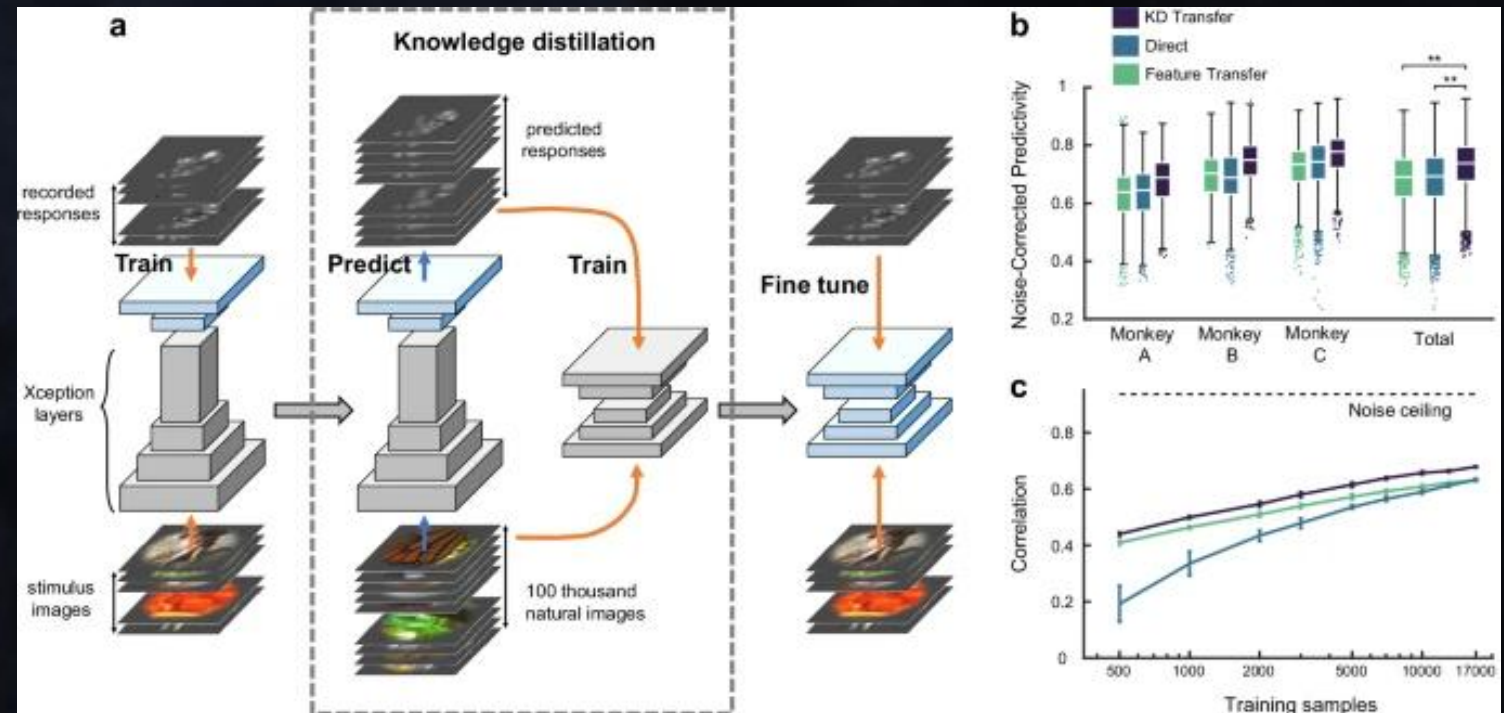


Fig. 2: DNN modeling on cortical response dataset from: Wang, T., Lee, T. S., Yao, H., Hong, J., Li, Y., Jiang, H., Andolina, I. M., & Tang, S. (2024). Large-scale calcium imaging reveals a systematic V4 map for encoding natural scenes. *Nature Communications*, 15(1). <https://doi.org/10.1038/s41467-024-50821-z>
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Exemplary Study in Primates

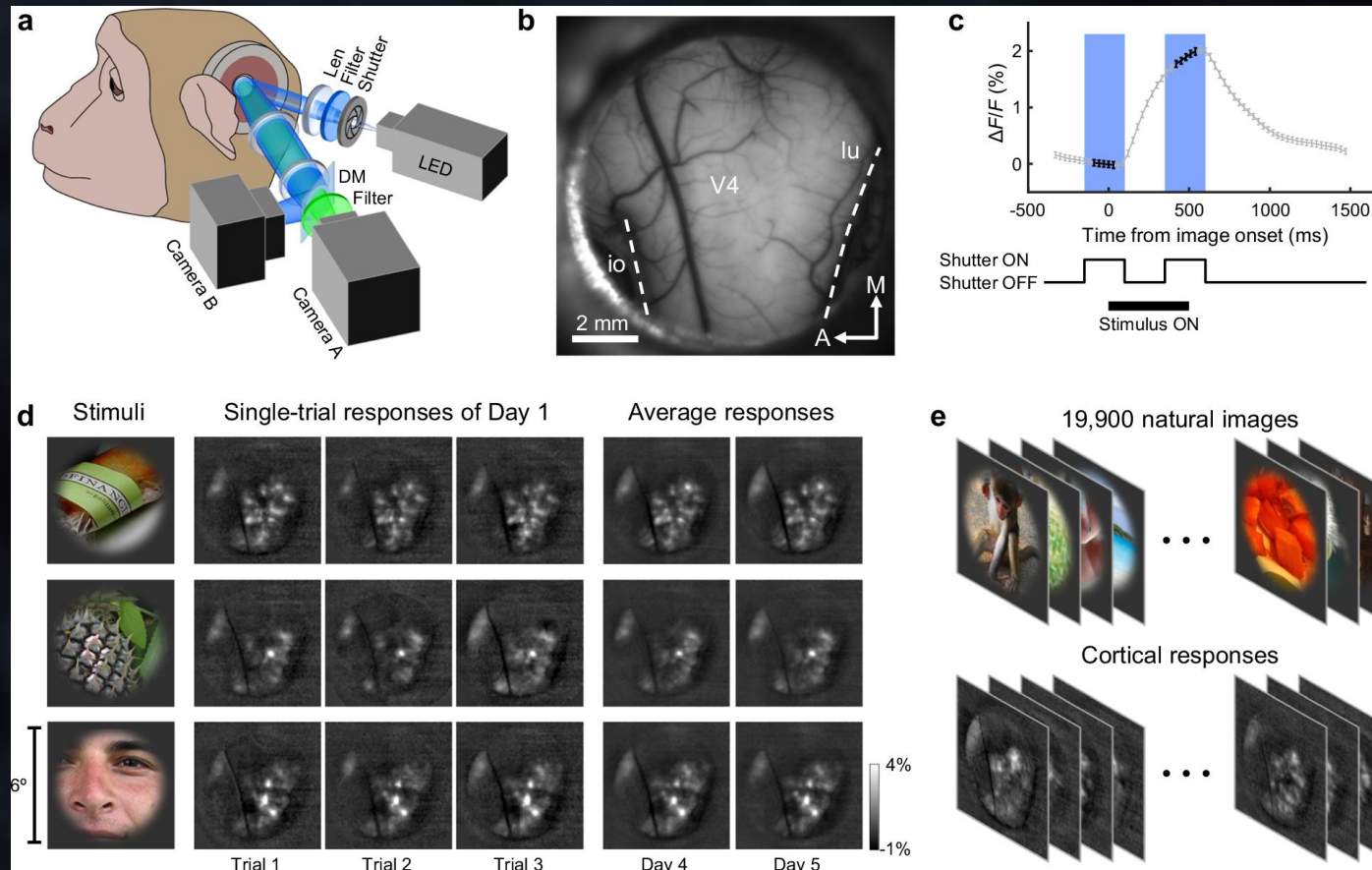


Fig. 1: Widefield calcium imaging of V4 cortical responses to natural scenes from: Wang, T., Lee, T. S., Yao, H., Hong, J., Li, Y., Jiang, H., Andolina, I. M., & Tang, S. (2024). Large-scale calcium imaging reveals a systematic V4 map for encoding natural scenes. *Nature Communications*, 15(1). <https://doi.org/10.1038/s41467-024-50821-z>
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Exemplary Study in Primates

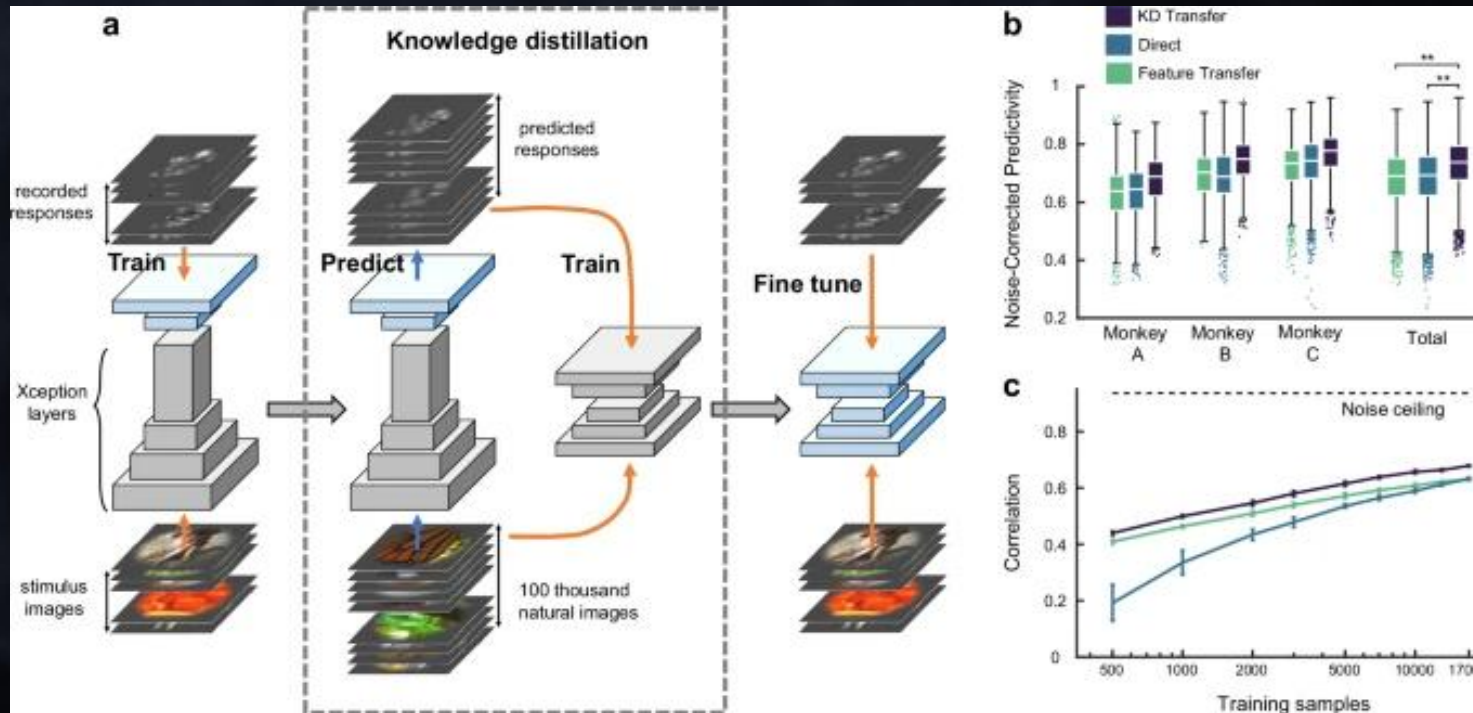


Fig. 2: DNN modeling on cortical response dataset from:
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Exemplary Study in Primates

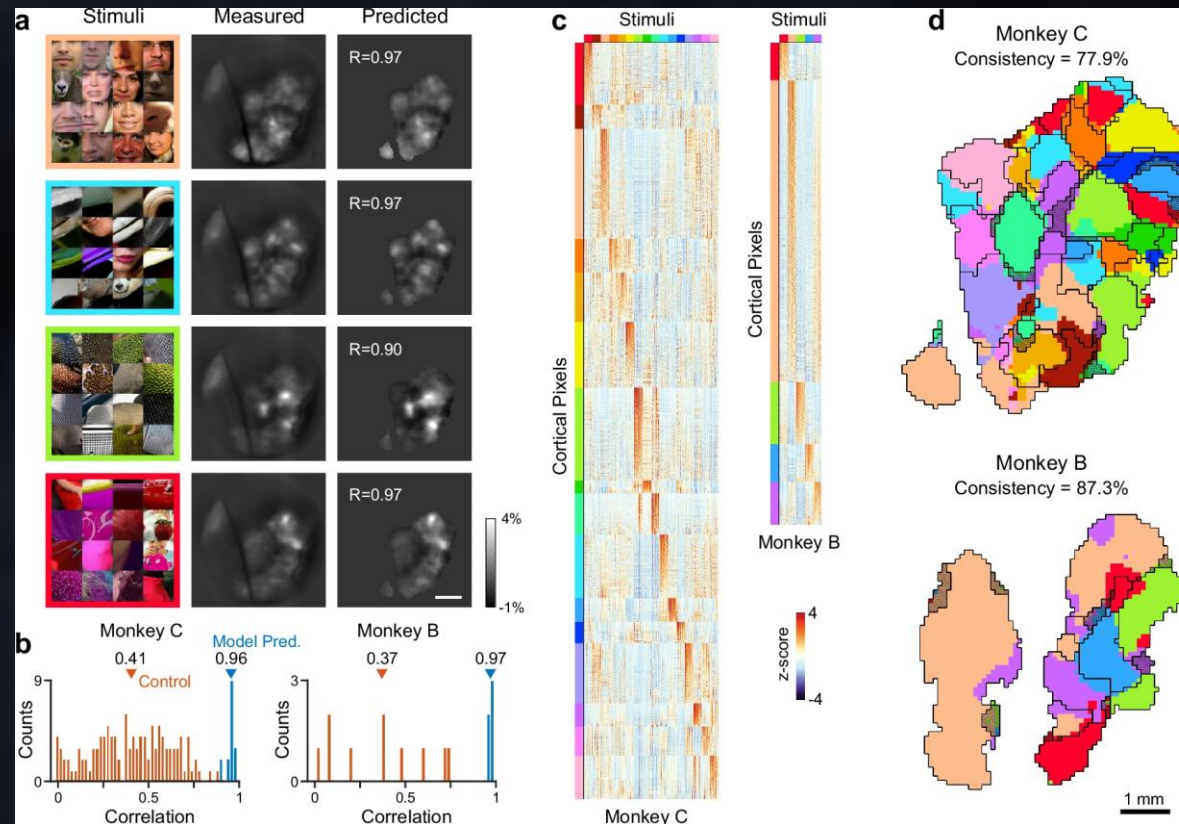


Fig. 4: Experimental verification of the model-predicted preference map.

Wang, T., Lee, T. S., Yao, H., Hong, J., Li, Y., Jiang, H., Andolina, I. M., & Tang, S. (2024). Large-scale calcium imaging reveals a systematic V4 map for encoding natural scenes. *Nature Communications*, 15(1).

<https://doi.org/10.1038/s41467-024-50821-z>

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6. Researchers

CONSCIOUSNESS RESEARCH NEEDS YOU



Image by [Vector Portal](#), [CC BY 4.0](#), via Wikimedia Commons¹

Future of consciousness research and concluding remarks

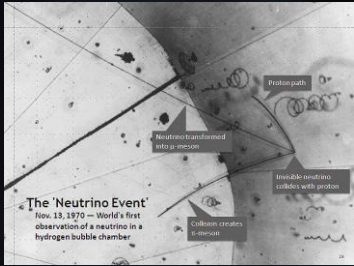


Image by [Argonne National Laboratory](#), Public domain, via Wikimedia Commons



Image by [Courtesy: National Human Genome Research Institute](#), Public domain, via Wikimedia Commons

Who are we?

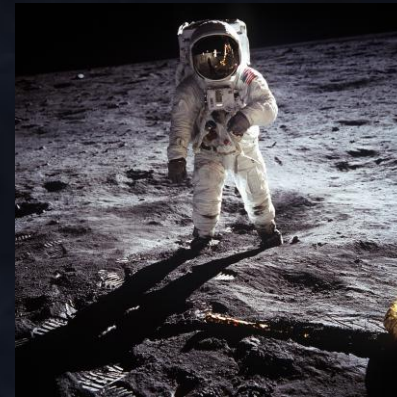


Image by [Neil A. Armstrong](#), Public domain, via Wikimedia Commons



Image by [NASA: ESA: G. Illingworth, D. Magee, and P. Oesch, University California, Santa Cruz; R. Bouwens, Leiden University; and the HUDF09](#), Public domain, via Wikimedia Commons

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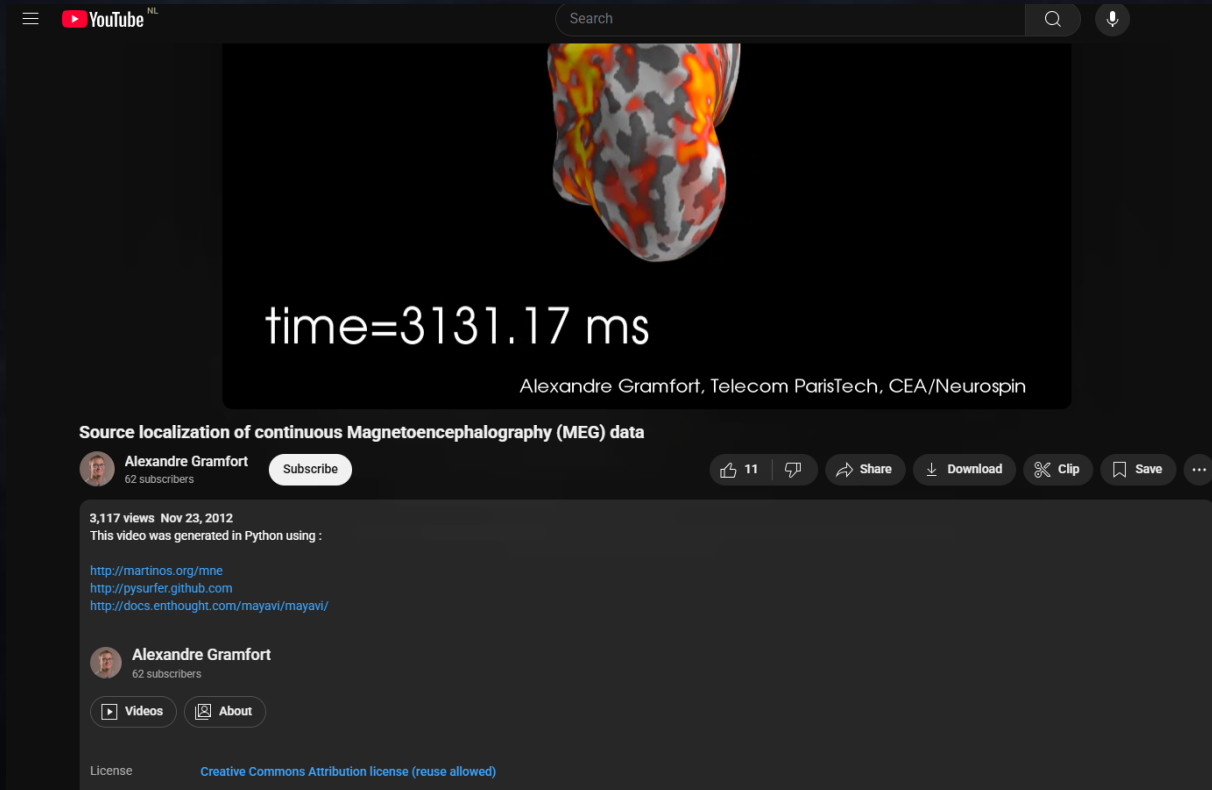
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Tests for consciousness in humans and beyond

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Highlights

Developing validated tests for consciousness (C-tests) applicable to many different systems is a key challenge for consciousness science. We suggest a framework for doing so, highlighting fundamental challenges with respect to validation.

We propose a four-dimensional space within which potential C-tests can be positioned.

We suggest that a promising strategy is to focus on non-trivial human cases (infants, fetuses, disorders of consciousness) and then progress toward nonhuman systems (animals, artificial intelligence, neural organoids).

C-tests can inform and shape our understanding of consciousness and our evaluation of theories of consciousness.

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Tests for consciousness in humans and beyond

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Large-scale two-photon calcium imaging in freely moving mice

[Weijian Zong](#)^{1,*}, [Horst A Obenhaus](#)¹, [Emilie R Skytøen](#)¹, [Hanna Eneqvist](#)¹, [Nienke L de Jong](#)¹, [Ruben Vale](#)¹, [Marina R Jorge](#)¹, [May-Britt Moser](#)¹, [Edvard I Moser](#)^{1,2,**}

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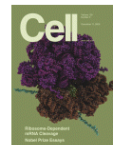
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Summary

We developed a miniaturized two-photon microscope (MINI2P) for fast, high-resolution, multiplane calcium imaging of over 1,000 neurons at a time in freely moving mice. With a microscope weight below 3 g and a highly flexible connection cable, MINI2P allowed stable imaging with no impediment of behavior in a variety of assays compared to untethered, unimplanted animals. The improved cell yield was achieved through a optical system design featuring an enlarged field of view (FOV) and a microtunable lens with increased z-scanning range and speed that allows fast and stable imaging of multiple interleaved planes, as well as 3D functional imaging. Successive imaging across multiple, adjacent FOVs enabled recordings from more than 10,000 neurons in the same animal. Large-scale proof-of-principle data were obtained from cell populations in visual cortex, medial entorhinal cortex, and hippocampus, revealing spatial tuning of cells in all areas.



Large-scale two-photon calcium imaging in freely moving mice

Author: Weijian Zong, Horst A. Obenhaus, Emilie R. Skytøen, Hanna Eneqvist, Nienke L. de Jong, Ruben Vale, Marina R. Jorge, May-Britt Moser, Edvard I. Moser

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Fast two-photon imaging of subcellular voltage dynamics in neuronal tissue with genetically encoded indicators

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Abstract

Monitoring voltage dynamics in defined neurons deep in the brain is critical for unraveling the function of neuronal circuits but is challenging due to the limited performance of existing tools. In particular, while genetically encoded voltage indicators have shown promise for optical detection of voltage transients, many indicators exhibit low sensitivity when imaged under two-photon illumination. Previous studies thus fell short of visualizing voltage dynamics in individual neurons in single ...

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Ethics

Animal experimentation: Animal experiments were performed in accordance with either (1) the recommendations in the Guide for the Care and Use of Laboratory Animals of the National Institutes of Health and the guidelines of the Stanford Institutional Animal Care and Use Committee under Protocol APLAC-23407, or (2) the guidelines for animal welfare of the Canadian Council on Animal Care and protocols approved by the Université Laval Animal Protection Committee (protocol number 2014-149-3). All surgery was performed under sodium pentobarbital anesthesia, and every effort was made to minimize suffering.

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